

Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48×10^{-8} cm
B. 5.92×10^7 cm/sec
C. 8.66×10^{-6} cm
D. 3.14×10^{-5} cm
E. 9.81×10^{-4} cm
F. 5.27×10^{-9} m
G. 1.23×10^{-7} cm
H. 4.14×10^{-10} m
I. 6.625×10^{-27} erg sec
J. 0.512×10^6 eV

Reasoning + Answer:

reasoning process here
The de Broglie wavelength λ of a particle is given by :

$$\lambda = h / p$$

where :
- h = Planck's constant = 6.626×10^{-34} J $\cdot$ s = 6.626×10^{-27} erg $\cdot$ s
- p = momentum of the particle

For a non-relativistic electron , kinetic energy $K = (1/2)mv^2$, and momentum $p = \sqrt{2mK}$

We can substitute to get :

$$\lambda = h / \sqrt{2mK}$$

We need to calculate λ for $K = 1$ eV = 1.602×10^{-19} J

The mass of an electron $m = 9.109 \times 10^{-31}$ kg

Now calculate p :

$$\begin{aligned} p &= \sqrt{2 \times 9.109 \times 10^{-31} \times 1.602 \times 10^{-19}} \\ &= \sqrt{2.918 \times 10^{-50}} \\ &\approx 5.402 \times 10^{-25} \text{ kg}\cdot\text{m/s} \end{aligned}$$

Now calculate λ :

$$\begin{aligned} \lambda &= (6.626 \times 10^{-34}) / (5.402 \times 10^{-25}) \\ &\approx 1.227 \times 10^{-9} \text{ m} \end{aligned}$$

Now convert to cm :
 $1 \text{ m} = 100 \text{ cm} \implies 1.227 \times 10^{-9} \text{ m} = 1.227 \times 10^{-7} \text{ cm}$

Now check the options in cm :

- A. 2.48×10^{-8} cm $\implies$ too small
- B. 5.92×10^7 cm/sec $\implies$ not a wavelength
- C. 8.66×10^{-6} cm $\implies$ too large
- D. 3.14×10^{-5} cm $\implies$ too large
- E. 9.81×10^{-4} cm $\implies$ too large
- F. 5.27×10^{-9} m $\implies$ is 5.27×10^{-11} cm $\implies$ not matching
- G. 1.23×10^{-7} cm $\implies$ matches 1.227×10^{-7} cm
- H. 4.14×10^{-10} m $\implies$ 4.14×10^{-12} cm $\implies$ no
- I. 6.625×10^{-27} erg sec $\implies$ not a wavelength
- J. 0.512×10^6 eV $\implies$ energy , not wavelength

So , the correct value is approximately 1.23×10^{-7} cm

< answer > G </answer >